Qihan Shan

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EDUCATION

University of Pennsylvania Philadelphia, PA May 2025 – May 2027 (expected)
M.S. in Mechanical Engineering and Applied Mechanics — Mechatronics & Robotic Systems
Coursework: Robot Motion Control, Path Planning, Robot Kinematics, Machine / Deep / Reinforcement Learning

Zhejiang University–UIUC Institute (ZJUI) Sep 2021 – Jun 2025
B.S. in Mechanical Engineering, minor in Electrical Engineering | GPA 3.87 / 4.00

EXPERIENCE

AgiBot — Motion Control Algorithm Intern Shenzhen, China Jun 2026 – Present
Central R&D, Motion Intelligence Group | Layered navigation stack for the X2 humanoid

Semantic Planning — Dual-System Vision-and-Language Navigation for a Humanoid

- Built the navigation data pipeline end to end: collected 4,229 trajectories across 60 Matterport3D scenes, projected the 3-D paths onto egocentric views with depth-based occlusion filtering, and segmented them into 251,916 training samples.
- LoRA-finetuned a 7B vision-language model to emit the farthest reachable waypoint in view as a pixel coordinate, and to turn or look down on its own when that point falls outside the frame. Freezing it, learnable latent queries pull its hidden state out as the condition for a flow-matching network generating 32 dense waypoints; the two run at 2 Hz and 30 Hz so slow reasoning never blocks control.
- VLN-CE RxR val-unseen: SR 59.3 / SPL 51.6. Projecting those goals into body-frame waypoints for the local planner closed the path from a language instruction to motion on the real X2.

Local Execution — Perceptive Forward-Dynamics Navigation on a Humanoid

- Built a local planner around a learned dynamics model. Rather than regressing poses, it predicts the residual between commanded and achieved velocity and integrates that into 5 s of future motion, with a separate head emitting a per-step failure probability; an MPPI planner then scores candidates on distance to goal and predicted risk alone, with no hand-tuned cost maps.
- Deployed the full perception–planning–control pipeline on hardware, clearing doorways, pillar fields and corridors on flat ground and reaching goals up to 10 m away within 0.3 m.
- Built a 10 Hz perception–replanning loop for moving obstacles, accumulating risk across neighbouring sampled paths so one missed prediction cannot clear a whole fan of commands: 100% avoidance in simulation against a 1 m/s obstacle within 4 m, and 80% on hardware where sensing-to-actuation latency bites, the other 20% holding position rather than colliding — no collisions in either setting.

Perceptive Locomotion — Attention-Based Perceptive Locomotion for a Humanoid

- Trained an attention-based map-encoding policy: a query assembled from global map features and proprioception weights the local features, so the policy attends to the ground under the next foothold. The map carries elevation and per-cell uncertainty as separate channels, making an occlusion read as unknown rather than as flat. A privileged-map teacher is then distilled into a student running on nothing but the map it builds itself.
- Crossed stepping stones, stairs and slopes in both directions, and discrete raised platforms, arriving at each goal on the commanded heading: 90% teacher / 75% student success on unseen terrain.
- Set termination conditions and the terrain curriculum for an armed humanoid; diagnosed and fixed foothold drift on consecutive stair ascents traced to odometry–elevation-map misalignment.

PROJECTS

UAV 3-D Path Planning and Real-Flight Deployment Feb – May 2026
UPenn | Advisor: Prof. M. Ani Hsieh

- Designed an SE(3) pose controller with a low-level PD attitude loop and combined A* global planning with piecewise quintic-polynomial fitting for smooth, collision-free, dynamically feasible trajectories; integrated the full planning-to-control stack on hardware and flew autonomously through a real 3-D maze to a commanded landing point.

Quadruped Loco-Manipulation with a Mounted Arm Sep 2024 – Jan 2025
ZJU-UIUC Physical Intelligence Lab | Advisor: Prof. Hua Chen

- Led whole-body control modeling for dynamic mobile grasping — body posture, foot contacts, end-effector pose and stability in a single policy — and trained PPO policies in massively parallel simulation with rewards spanning locomotion, manipulation and their coordination.

SKILLS

Programming: Python, C, C++, MATLAB
Learning & Simulation: PyTorch, Isaac Lab / Isaac Sim, Habitat, MuJoCo; PPO, LoRA/PEFT, teacher–student distillation, diffusion policy and flow matching
Algorithms & Deployment: MPPI sampling-based planning, learned forward dynamics models, elevation mapping, VLN data pipelines, sim-to-real onboard deployment
Tools: ROS2, Linux, Git, SolidWorks, KiCad
Awards & Languages: Second Prize, National RoboMaster Robotics Competition 3v3, 2023 (ZJUI Meta RoboMaster Team) | Mandarin (native), English (TOEFL 103)